Comparison of Transfer Learning and Traditional Machine Learning Approach for Text Classification

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***Abstract*— In the era of big data where data consist of images, text, audio etc., manually analysing text is a big challenge. Over a long time, research in Natural Language Processing (NLP) has relied upon rule based approach and machine learning approach but these approaches need domain expert to build only task specific NLP systems. However, the recent research in NLP has given rise to pre-trained language models like ELMo, GPT, BERT, DistilBERT etc. which have made transfer learning as state-of-the-art technique in NLP. In this paper, we present the comparison of transfer learning and traditional machine learning approach for text classification. To achieve our goal, we have fine-tuned BERT and DistilBERT for text classification to compare transfer learning approach against the traditional machine learning approach based on Term Frequency-Inverse Document Frequency (TF-IDF) algorithm for feature extraction. The performance of all models is empirically tested on suite of different datasets. The results obtained by fine-tuning pre-trained language models show significant improvement over traditional machine learning approach for text classification. This research study also promotes the use of DistilBERT over BERT as default transfer learning model for various text classification applications because it can be trained in less time with less number of parameters to give similar performance like BERT.**

Keywords—text classification, natural language processing, machine learning, transfer learning, pre-trained models

I. Introduction

Text classification is a sub-field of Natural Language Processing (NLP). It is used in many applications like news categorization, sentiment analysis [1], spam detection, recommendation systems [2] etc. [3]. In the review of literature, it is observed that different approaches like rule based approach [4], statistical approach [5] and neural approach have been used to solve this task. Rule based approach relies on manual writing of rules for building NLP systems and statistical approach also known as machine learning approach is based on feature engineering process in which we extract various features from text by using statistical methods. The limitation of both approaches is that they need domain expert for each NLP task to build task specific model only. However, this problem is addressed by neural approach which is based on neural networks and recently became more popular to solve NLP tasks using deep learning. Automatic feature engineering is the key advantage of deep neural networks over non neural methods used earlier for solving NLP tasks. Various deep learning based neural networks like convolution neural networks [6-7], recurrent neural networks [8-9], graph neural networks [10-12] and attention mechanism [13-14] are applied in the field of text classification [15].

Deep learning based pre-trained language models (PLMs) like ELMo [16], GPT [17], BERT [18] etc. have given rise to a technique called transfer learning (TL) in NLP. It is a process in which knowledge acquired while training a model on a particular task or domain can be reused to solve downstream task or domain [19]. There are many NLP tasks and domains for which it is very difficult to find large scaled labelled data. PLMs are pre-trained on one or more tasks and can be reused to solve several other NLP tasks using the process of fine-tuning as shown in Fig. 1. Unlike machine learning, TL eliminates the need to build task specific language models.

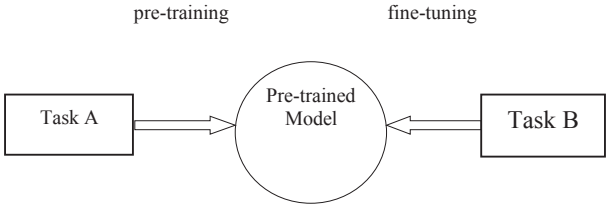

Fig. 1. Transfer learning setup

LMs are pre-trained with huge amount of unlabelled text and can be fine-tuned with small amount of labelled data to perform various NLP tasks [15]. Dai and Le [20] first proposed the idea to pre-train language models. Among various PLMs, BERT is considered as state-of-the-art (SOTA) for TL due to its ability to predict context of the word from both directions, left and right [18]. However, BERT's model size is very large and many research studies have been conducted in the literature to find optimized variants of BERT like DistilBERT [21], ALBERT [22], RoBERTa [23], TinyBERT [24] etc. Among these, DistilBERT is smaller, cheaper and faster version of BERT. Therefore we have selected BERT and its optimized version DistilBERT to compare transfer learning against classical machine learning approach based on Term Frequency - Inverse Document Frequency (TF-IDF) algorithm which is used to extract features from text. Transfer learning has shown remarkable results in computer vision [25] and now a days it is gaining momentum in the field of NLP as well. The motivation behind this research is to explore the potential of TL in NLP against traditional machine learning.

The comparisons are made on the basis of experimental study on four different datasets used for text classification.

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In each experiment, different text classifiers are built using BERT, DistilBERT and TF-IDF based machine learning approach.

The rest of this paper is organized as follows. In section 2, related work is discussed. In section 3, methodology used to perform comparative study is described. In section 4, datasets, tools & techniques used, and experimental details are given. In section 5, results obtained in the experiments are presented. In section 6, conclusions are made along with future scope of work in section 7.

II. Related Work

During recent years, a lot of research has been carried out to build efficient PLMs to solve NLP tasks. Among various PLMs, BERT has achieved SOTA results in wide range of NLP tasks [18]. Soon after its release, researchers have started to adopt BERT for sentiment analysis [26], question answering [27], text summarization [28] and document retrieval [29]. One of the major limitations of BERT is large model size which leads to difficulty in pre-training BERT model from scratch. Researchers have found various methods to compress the model size and have developed several optimized variants of BERT like DistilBERT [21], ALBERT [22] , RoBERTa [23] etc. All these models can be used to solve NLP tasks using TL.

The performance of machine learning models tends to decrease while processing large amount of text. With the rise of high computation resources like GPU, TPU etc., deep learning models are pre-trained on huge amount of unlabeled text and can be fine-tuned with small amount of data to solve various NLP tasks using TL. So the question that is discussed here is to compare the efficiency of transfer learning technique with that of classical NLP technique.

Tenney, Das and Pavlick [30] found that the steps of traditional NLP pipeline can be represented by BERT and it can also adjust the pipeline dynamically by updating lower level decisions based on information disambiguation from higher-level representations. Usherwood and Smit [31] performed comparison of BERT and ULMFiT with classical NLP approaches in low-shot classification tasks and concluded BERT as SOTA pre-trained model for NLP. A similar work by González-Carvajal and Garrido-Merchán [32] shows the superiority of BERT over classical approaches on different datasets.

However, many comparative studies in the literature have focused on BERT as default transfer learning model. In this work, we have introduced DistilBERT along with BERT as SOTA TL models and compared both the models against traditional machine learning approach. A comparative analysis is also made between BERT and DistilBERT in terms of training parameters and model accuracy based on experimental study and review of literature.

III. Methodology Used

This paper presents the comparison between traditional machine learning approach and transfer learning approach for text classification. The following subsections will briefly describe the methodology used in both the approaches.

A. Machine Learning Approach

In this paper, we have used the traditional machine learning approach based on TF-IDF algorithm. TF-IDF algorithm is used to build bag-of-words model in classical NLP. It is used to extract features from text which are then fed to machine learning algorithm to build a classifier. It comprises of two terms: Term Frequency (TF) and Inverse Document Frequency (IDF). TF calculates the frequency of a term in the document. Given a term t_i in a document, its term frequency can be denoted by tf_i. If N documents exist in a corpus and the term t_i occurs in n_i of these documents. Then, IDF can be calculated using (1) [33]

$$idf(t_i) = logN/n_i \tag{1}$$

Finally, TF-IDF for a given term t_i can be calculated using (2) [33]

$$tfidf(t_i) = tf_i * idf(t_i) \tag{2}$$

Fig. 2 represents the machine learning approach for text classification used in this paper.

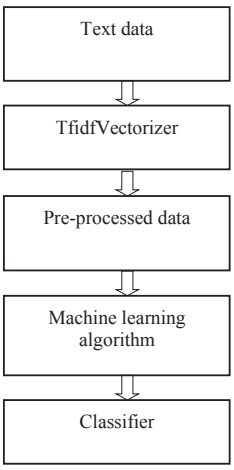
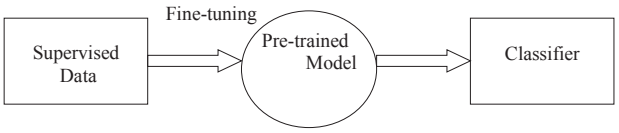


Fig. 2. Machine learning approach

B. Transfer Learning Approach

PLMs have given rise to transfer learning approach in NLP. In this approach, we have used a small amount of supervised data to fine-tune a PLM for building text classifier. Fig. 3 represents the transfer learning approach using pre-trained model.



Transfer learning approach

This approach eliminates the need to build a model from scratch for each NLP task. There are many PLMs available in the literature but we have selected SOTA BERT and its optimized version DistilBERT for comparison. The following subsections will provide a brief description of both models.

1) BERT: BERT is a PLM developed by Google researchers. The model was built in two phases: pre-training

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and fine-tuning. Pre-training of model is performed with two tasks: Masked Language Model and the Next Sentence Prediction on BooksCorpus and English Wikipedia. BERT exists in two sizes as BERTBASE (110 million) and BERTLARGE (340 million) depending upon the number of parameters on which model is trained [18]. "It can also be fine-tuned by adding one output layer to pre-trained model architecture specific to each task without major modifications in original architecture to create SOTA models for wide range of NLP tasks like text classification, language inference, question answering etc. BERT has obtained SOTA results on 11 NLP tasks including GLUE [34] score improved to 80.5" [35]. It is based on Transformer [14] which uses self-attention mechanism to capture the relationships between words.

2) DistilBERT: DistilBERT is an optimized version of BERT. It's smaller, lighter and faster than BERT and roughly matches its performance [21]. The process of knowledge distillation used during the pre-training phase reduces the BERT's model size by 40%, while keeping 99% of its language understanding capabilities and being 60% faster. "A triple loss combining language modeling, distillation and cosine-distance losses are used in the model to leverage the inductive biases learned by larger models during pre-training" [21]. The model can be trained with less number of parameters as compared to other PLMs.

IV. Experiments

In this section, we begin with a brief description of datasets used for text classification, and then we have mentioned various tools and techniques used to perform experiments followed with implementation details of models for both approaches under different settings.

A. Datasets

In this paper, we have used four different text classification datasets to perform comparative study. The description of datasets is given as follows:

1) Disaster Tweet Prediction: Disaster Tweet Prediction dataset is downloaded from kaggle website [36]. It is comprised of 7613 rows in train data file and 3263 rows in test data file. The task is to build a classifier that can classify disaster tweet as real or fake.

2) SMS Spam Collection: SMS Spam Collection is a popular dataset used for binary text classification [37]. This dataset has 4825 ham and 747 spam messages. It is an imbalanced dataset due to uneven class distribution.

3) 20 Newsgroups: 20 Newsgroups dataset consists of newsgroup documents on 20 different categories [38]. We have used a subset of 20 Newsgroups dataset comprising 5 categories only to perform multi-class text classification.

4) IMDB: IMDB dataset is a collection of 50,000 movie reviews. It is uniformly distributed into train and test set with 25,000 rows in each set [39]. Here, the task is to perform binary sentiment classification of movie reviews as positive or negative.

Table I presents the brief summary of text classification datasets described above.

TABLE I. Dataset Description

Datasets	Task	Categories	Description
Disaster Tweet Prediction	Binary Classification	real (1) fake (0)	To classify tweets as real or fake
SMS Spam Collection	Binary Classification	ham spam	To classify messages as ham or spam
20 Newsgroups	Multi-class classification	alt.atheism comp.graphics sci.electronics sci.med rec.sport.hockey	To classify news into one of these five categories
IMDB	Binary Classification	positive negative	To perform sentiment classification of movie reviews

B. Tools and Techniques

For machine learning, we have used H2OAutoML from python 3 h2o module to perform binary text classification for Disaster Tweet Prediction and SMS Spam Collection datasets. h2o module is an open source python package that is used to build and test machine learning models rapidly. The various steps used in a machine learning pipeline like data pre-processing using TF-IDF, hyper-parameter tuning and selection of optimal machine learning algorithm for a dataset can be done automatically using H2O's AutoML function. In case of 20 Newsgroups and IMDB datasets we have used TFidfVectorizer from sklearn python 3 package to preprocess the dataset. Pre-processed dataset is then used to train various machine learning algorithms to perform text classification.

For transfer learning, we have used ktrain python 3 package [40] to fine-tune BERT and DistilBERT models on different datasets. ktrain is an open source python module that wraps keras and tensorflow and used to train neural network in an efficient manner.

C. Models and Settings

We have built various models under different technical settings for each dataset to perform the comparative study between two approaches for text classification. The details of experiments conducted for each dataset are given in this subsection.

1) Disaster Tweet Prediction: In the first experiment, training data is divided into training (80%) and validation set (20%). Training set is then used to fine-tune BERT and DistilBERT for 2 epochs with learning rate 1e-5. For DistilBERT, batch size was set to 6 with maxlen 500. In case of BERT, batch size was set to 16 and maxlen 65. Batch size and maxlen can be set according to reference range given in ktrain module [40]. Both the models are tested on validation set.

For machine learning we have used H2OAutoML from h2o module which gives best classifier with model_id StackedEnsemble_BestOfFamily_AutoML_20210624_113923 with accuracy 0.69.

2) SMS Spam Collection: SMS Spam Collection is an imbalanced dataset because it has 4825 ham and 747 spam messages. In this experiment, we first balanced the dataset. The balanced dataset is comprised of 747 ham messages and

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747 spam messages. So there are total 1494 rows in our dataset. After this, data is split into train (80%) and validation set (20%). BERT and DistilBERT models are fine-tuned using training set for 2 epochs with learning rate 1e-5. The values for batch size and maxlen for both models are set as in the first experiment. These models are tested with validation set.

For machine learning approach, H2OAutoML from h2o module gives best classifier with model_id DRF_1_AutoML_20210626_132547 with accuracy 0.94.

3) 20 Newsgroups: In the third experiment, we have used a subset of 20 Newsgroups dataset comprising of 5 categories (alt.atheism, comp.graphics, sci.electronics, sci.med, rec.sport.hockey) to perform multi-class text classification. Here, training data consists of 2849 rows and validation data consists of 1896 rows.

Firstly, dataset is loaded from sklearn python 3 module and pre-processed using TF-IDF algorithm for machine learning and trained on popular multi-class text classification algorithms like Naïve Bayes, Linear SVC, Logistic Regression (LR) and Random Forest Classifier (RFC).

For transfer learning, BERT and DistilBERT models are fine-tuned on the same dataset for 5 epochs with learning rate 2e-5 and batch size 16.

4) IMDB: For machine learning, we have pre-processed the data with TF-IDF and it was trained on various machine learning algorithms like Multinomial Naïve Bayes, Passive Aggressive Classifier (PAC), Ridge Classifier, Linear SVC, LR and Voting Classifier.

For transfer learning, training data is used to fine-tune BERT for 1 epoch with learning rate 2e-5 and DistilBERT for 2 epochs with learning rate 2e-5 .Test data is used for validating these models. Here BERT took one hour to get trained for 1 epoch only while DistilBERT was trained within 5 minutes for 2 epochs.

V. Results And Discussion

The results of the comparative study based on experiments conducted on different datasets are summarized in Table II and Table III. It can be observed from Table II that BERT and DistilBERT are giving outstanding performance when fine-tuned with very less amount of data as compared to traditional machine learning models.

TABLE II. ACCURACY OF TRANSFER LEARNING AND MACHINE LEARNING MODELS ON DIFFERENT DATASETS

Method	Disaster Tweet Prediction	SMS Spam Collection	20 Newsgroups	IMDB
DistilBERT	0.84	0.98	0.96	0.98
BERT	0.84	0.98	0.96	0.97
H2O AutoML	0.69	0.94	-	-
Naïve Bayes	-	-	0.97	0.87
Linear SVC	-	-	0.96	0.89
LR	-	-	0.95	0.88
RFC	-	-	0.89	-
PAC	-	-	-	0.88
Ridge Classifier	-	-	-	0.89
Voting Classifier	-	-	-	0.89

Table III shows the comparative analysis of BERT and DistilBERT on the basis of training parameters and model accuracy. We have kept the same learning rate for both pre-trained models during the process of fine-tuning models for each dataset.

In each experiment, it can be observed that DistilBERT can be trained with less number of training parameters and gives similar performance like BERT. This observation reveals that DistilBERT can be preferred over BERT and it can be used as default TL model for different applications of text classification.

TABLE III. COMPARATIVE ANALYSIS OF BERT AND DISTILBERT

Dataset	DistilBERT				BERT			
	Model Size (#Parameters)	*Learning Rate*	*#Epochs*	*Accuracy*	*Model Size (#Parameters)*	*Learning Rate*	*#Epochs*	*Accuracy*
Disaster Tweet Prediction	66,955,010	1e-5	2	0.84	109,140,482	1e-5	2	0.84
SMS Spam Collection	66,955,010	1e-5	2	0.98	109,140,482	1e-5	2	0.98
20 Newsgroups	66,957,317	2e-5	5	0.96	109,486,085	2e-5	5	0.96
IMDB	66,955,010	2e-5	2	0.98	109,397,762	2e-5	1	0.97

VI. Conclusion

In this paper, we introduced the classical NLP approach based on TF-IDF and briefly described the pre-trained models like BERT and DistilBERT. A comparison was made between transfer learning and traditional machine learning approach based on four different experiments adding empirical evidence to state that transfer learning method is likely to give better results than traditional method. We found that the implementation of PLMs is far less complicated than classical NLP methods. This research study revealed that pre-trained models can be fine-tuned to give SOTA performance using very less amount of labelled data as compared to traditional machine learning models. Our work aimed to study the efficiency of TL in NLP. BERT has its own limitations in terms of model size, so we used DistilBERT (optimized version of BERT) to compare the results with BERT. From the experiments, it can also be concluded that DistilBERT can be used as default TL model for text classification.

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VII. Future Scope

In this study, learning rates are arbitrarily chosen for BERT and DistilBERT, however their results can be improved further by optimal selection of learning rates or hyper-parameter tuning. Existing PLMs are highly parameterized and very large in size which makes their use more expensive in terms of computational requirements. Due to this a lot of research is going on in the direction to improve PLMs for TL. DeBERTa [41] is the upcoming SOTA language model for TL. The results in the experiments can also be improved with the upcoming research in PLMs for NLP.

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